

LGBTI National Policy Dataset

Codebook

Version 2.2 — Long Format

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1. Dataset Overview

The LGBTI National Policy Dataset (Version 2.2) is a country–year–policy panel tracking the adoption, scope, and enforcement of LGBTI-related policies across 195 countries from 1991 to 2025. The dataset is distributed in long format: each row records one country in one year for one specific Law × Status combination.

This dataset and its composite index are an extension of the LGBTI Policy Index published in Velasco, Kristopher. 2023. "Transnational Backlash and the Deinstitutionalization of Liberal Norms: LGBT+ Rights in a Contested World." *American Journal of Sociology* 128(5): 1381–1429.

The dataset covers 27 distinct policy areas organized into 54 Law × Status combinations. Of these, 37 contain substantive data (at least one country with Proportion > 0 in some year); the remaining 17 are structural placeholders carrying all-zero values that keep the panel fully balanced for econometric methods.

While the panel starts in 1991, the temporal metadata columns (Year_of_National_Adoption, Year_National_Implementation) record full policy histories, so users can extend series back in time according to their own purposes.

1.1 Key features

- Balanced panel: 368,550 observations (195 countries × 35 years × 54 Law × Status combinations).
- 27 policy areas spanning criminalization, relationship recognition, anti-discrimination, hate-crime law, military service, gender identity, healthcare access, and constitutional protections.
- Population-weighted subnational scoring using constant 2000/2001 census shares for federal and devolved countries.
- Five time-varying indicators per active row drive the confirmatory factor analysis (CFA) pipeline: Population Coverage, Gender Scope, Penalty Severity, Evidence of Enforcement, and Ease of Access. Evidence_of_Enforcement is stored directly as a column; Sexes_Covered stores the numeric Gender Scope value; the other three are computed at runtime by the R script from the dataset's raw fields.
- Scope_Detail column carrying controlled per-area vocabularies that grade Ease of Access within policy areas that have a genuine within-policy access gradient (§4.5, §5.1.5).
- Twelve source-citation slots per group (Source_1 ... Source_12), with Source_1 holding the primary statute / case / decree citation.
- Complete value-level traceability: every cell in every row is attributable to one of the group's twelve source-citation slots (§6.10).
- Per-group data_confidence rating (high / medium / low) reflecting source-triangulation quality.
- Per-row row_type marker (substantive / structural_placeholder) for explicit panel partitioning.

1.2 Suggested citation

Velasco, Kristopher. 2026. "LGBTI National Policy Dataset, Version 2.2."
<https://doi.org/10.17605/OSF.IO/WT9P>. Distributed under CC BY 4.0.

1.3 Companion files

- LGBTI_Policy_Data_Long.csv — the longitudinal policy dataset (this codebook documents it).
- LGBTI_Composite_Index.csv — pre-built composite and unweighted index time series (§5, §7).
- Velasco_LGBTI_Index_Construction.R — CFA pipeline that regenerates the composite index from the panel CSV.
- LICENSE — full text of the Creative Commons Attribution 4.0 International License.

2. Quick-Start Guide

The following R code demonstrates how to load, explore, and analyze the dataset.

```
library(dplyr)
library(readr)
library(ggplot2)

# Load dataset
lgbti <- read_csv("LGBTI_Policy_Data_Long.csv")

# Restrict to substantive rows (skip the structural placeholders)
substantive <- lgbti %>% filter(row_type == "substantive")

# Restrict to active rows
active <- lgbti %>% filter(Proportion > 0)

# Restrict to high-confidence groups only
hi_conf <- lgbti %>% filter(data_confidence == "high")

# Example: Marriage Equality adoption over time
me_trend <- lgbti %>%
  filter(Law == "Marriage Equality", Status == "Legal") %>%
  group_by(Year) %>%
  summarise(n_countries = sum(Proportion > 0))

ggplot(me_trend, aes(Year, n_countries)) +
  geom_line() +
  labs(title = "Countries with Marriage Equality", y = "Number of Countries")
```

Stata users can load the CSV with ``import delimited``. Python users should use ``pandas.read_csv()``.

3. Dataset Structure

3.1 File format and dimensions

Single CSV file in long format: 368,550 rows × 36 columns. UTF-8 encoded.

3.2 Panel balance

The dataset is a fully balanced panel. Every country has exactly 1,890 rows (54 Law × Status combinations × 35 years). For combinations where a country has no relevant policy, all Proportion values are 0. Some structural-placeholder groups still carry source-slot text when the coding decision itself is documentary (e.g., a note establishing that no such statute has ever existed, or that a proposed instrument was never adopted); other placeholder groups leave the metadata columns empty.

The Year column indexes the panel year of observation; it is independent of Year_of_National_Adoption. A row's Year value carries no information about when the underlying statute was adopted: most rows have Year $\neq$ YNA because the establishing statute was signed before the panel window opened in 1991, or because the statute remained unchanged across the panel year. Year_of_National_Adoption holds the same value across every year-row of a given (Country, Law, Status) group.

3.3 Substantive vs. structural-placeholder rows

Every row carries a row_type value:

- substantive — the row belongs to a (Country, Law, Status) group with at least one year where Proportion > 0. 56,175 rows across 1,605 groups.
- structural_placeholder — the group is all-zero across 1991–2025. 312,375 rows across 8,925 groups. Kept for panel balance only.

At the (Law, Status) combination level: 37 substantive combinations versus 17 globally-all-zero structural placeholders.

3.4 Policy–status pairing logic

Each of the 27 policy areas is paired with exactly two status values representing opposing legal directions. Simultaneous positive proportions for both statuses occur in three legitimate cases: federal subnational splits, policy layering (coverage-then-restriction transitions in Gender Affirming Care), and single-year transition overlaps. Outside these three cases, paired statuses do not overlap for more than a single transition year (§6.13).

4. Variable Descriptions

The 36 columns are organized into seven groups, in this fixed order:

#	Column	Type	Group
1	Country	string	Identification
2	ISO	string	Identification
3	COW	numeric	Identification
4	Year	integer	Identification
5	Law	string	Policy
6	Status	string	Policy
7	Proportion	numeric [0,1]	Outcome
8	Year_of_National_Adoption	integer	Temporal
9	Year_of_First_Adoption	integer	Temporal
10	Year_of_National_Repeal	integer	Temporal
11	Blood_Restriction_Period	string	Supplementary
12	Max_Punishment	numeric/string	Supplementary
13	Fine_Amount	numeric/string	Supplementary
14	Sexes_Covered	numeric {0.50, 1.00}	Supplementary
15	Sub_National_Variation	binary 0/1	Supplementary
16	Sub_National_Coverage	string	Supplementary
17	Year_Law_Changed	integer	Temporal
18	Gender_Change_Requirement	string	Supplementary
19	Civil_Union_Benefits	numeric 1–5	Supplementary

20–27	Source_1 ... Source_8	string	Source
28	Year_National_Implementation	integer	Temporal
29	Evidence_of_Enforcement	five-anchor scale {0.00, 0.25, 0.50, 0.75, 1.00}	Indicator
30–33	Source_9 ... Source_12	string	Source
34	data_confidence	high/medium/low	Quality
35	row_type	substantive/structural_placeholder	Quality
36	Scope_Detail	string (controlled per-area vocabulary)	Supplementary

Note on indicators: of the five CFA indicators used to construct the composite index, only Evidence_of_Enforcement is stored as a dataset column. Sexes_Covered is the numeric Gender Scope indicator value. The remaining three indicators — Population Coverage, Penalty Severity, and Ease of Access — are computed at runtime by the R script (Velasco_LGBTI_Index_Construction.R) from the dataset's raw fields (Proportion, Max_Punishment, Scope_Detail, Gender_Change_Requirement, Civil_Union_Benefits, Blood_Restriction_Period).

4.1 Identification variables

Country, ISO (3-letter ISO 3166-1 alpha-3), COW (Correlates of War code), Year (1991–2025).

4.2 Policy variables

Law identifies one of 27 policy areas, organized into seven thematic domains. Each domain is illustrated below by a trend chart showing the population share of the world living under the policy each year. Blue shading marks progressive direction; red marks regressive.

4.2.1 Criminalization and Penalties

- Same-Sex Acts
- Age of Consent
- Death Penalty
- Morality/Propaganda
- LGBTQ+ Civil Society Restrictions

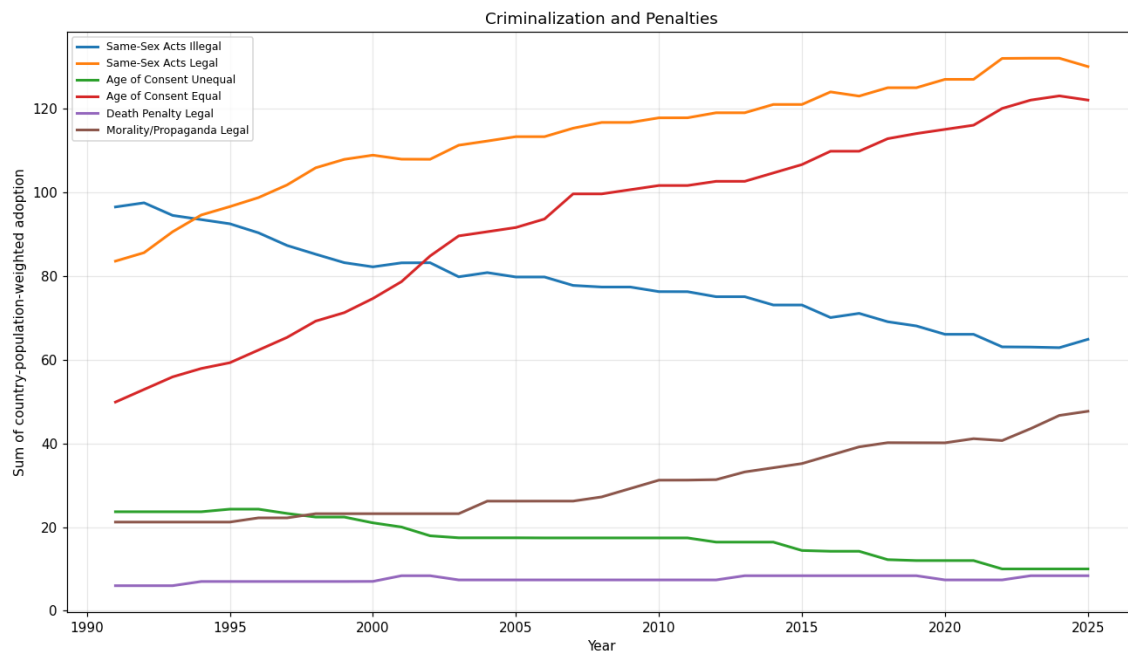


Figure 4.1 — Global trend in criminalization and penalty policies, 1991–2025.

4.2.2 Relationship Recognition

- Marriage Equality
- Civil Unions
- Joint Adoption

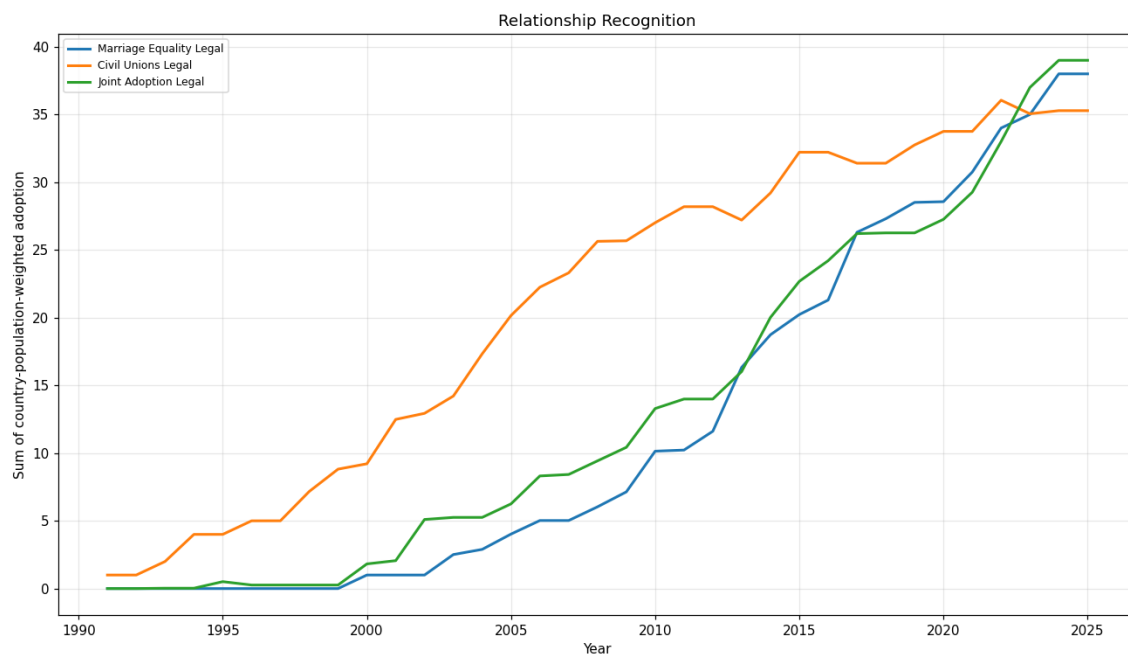


Figure 4.2 — Global trend in relationship-recognition policies, 1991–2025.

4.2.3 Anti-Discrimination Protections

- Employment Discrimination – Sexual Orientation
- Employment Discrimination – Gender Identity
- Goods/Services Discrimination – Sexual Orientation
- Goods/Services Discrimination – Gender Identity
- Conversion Therapies
- Religious Exemption Laws
- Gender Assignment Surgeries on Children

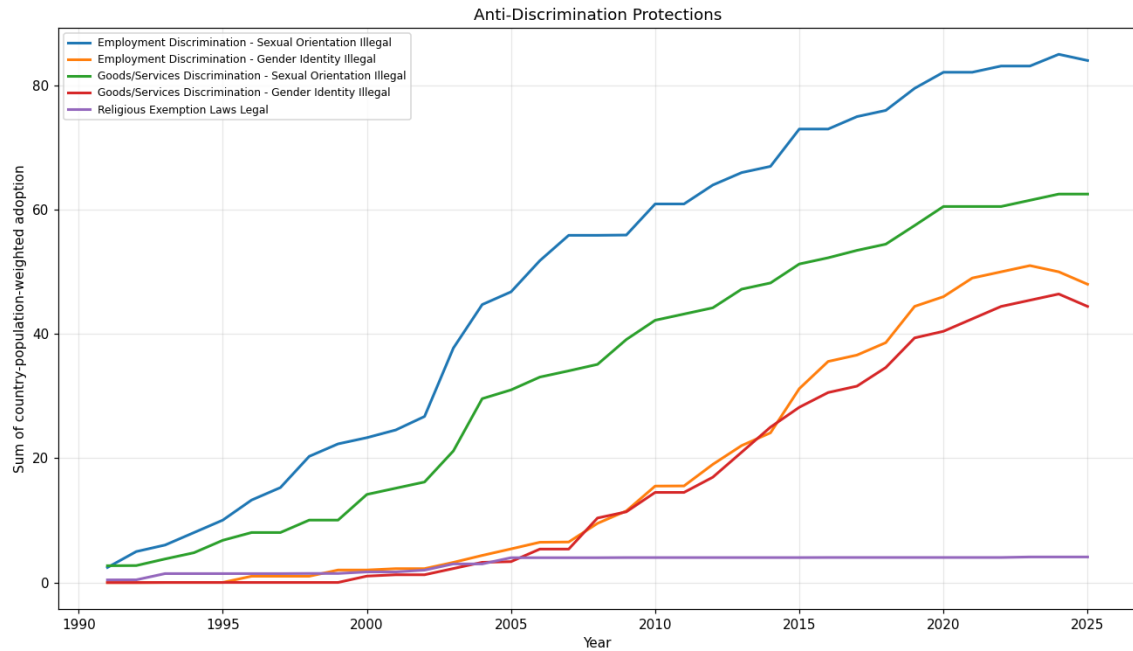


Figure 4.3 — Global trend in anti-discrimination protections, 1991–2025.

4.2.4 Hate Crime and Speech

- Hate Crime Protections – Sexual Orientation
- Hate Crime Protections – Gender Identity
- Incitement to Hatred

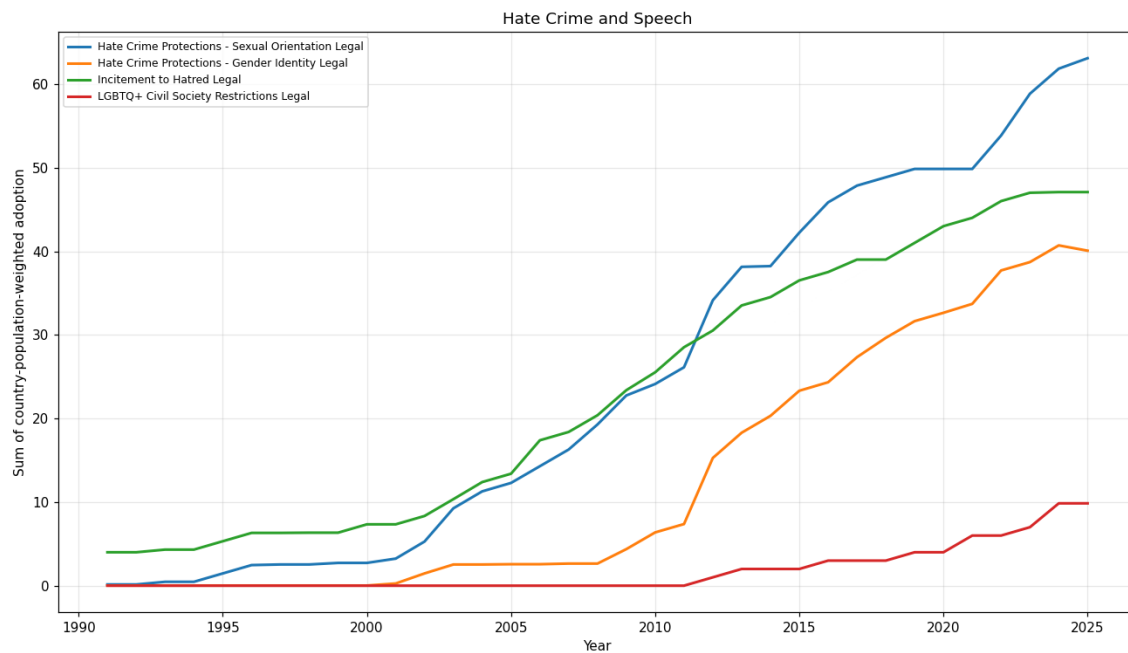


Figure 4.4 — Global trend in hate-crime and speech protections, 1991–2025.

4.2.5 Military Service

- LGB Military
- Transgender Military

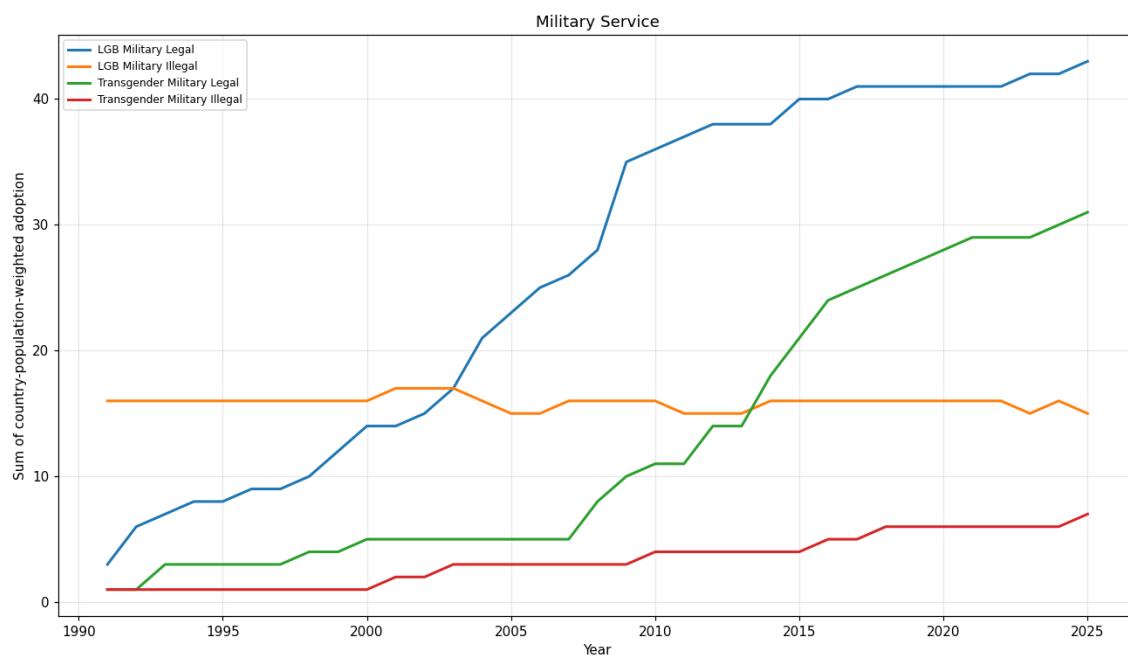


Figure 4.5 — Global trend in military-service policies, 1991–2025.

4.2.6 Gender Identity and Healthcare

- Gender Marker Change
- Third-Gender Recognition
- Gender Affirming Care – Adults
- Gender Affirming Care – Minors
- Blood Donations

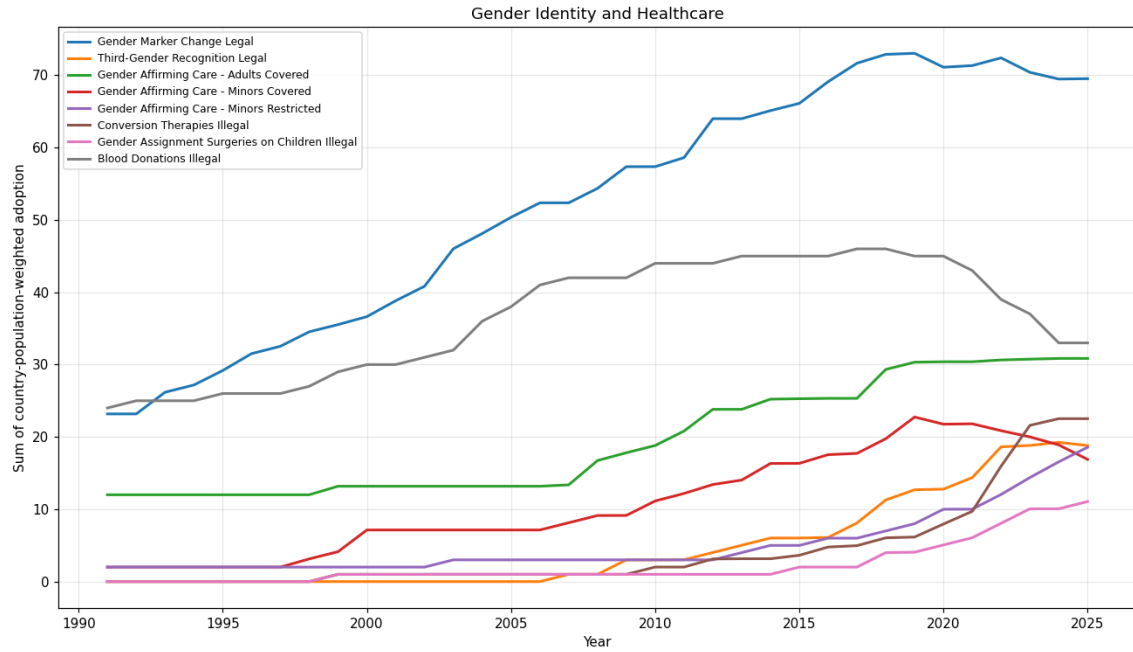


Figure 4.6 — Global trend in gender-identity and healthcare-access policies, 1991–2025.

4.2.7 Constitutional Protections

- Constitutional Protections – Sexual Orientation
- Constitutional Protections – Gender Identity

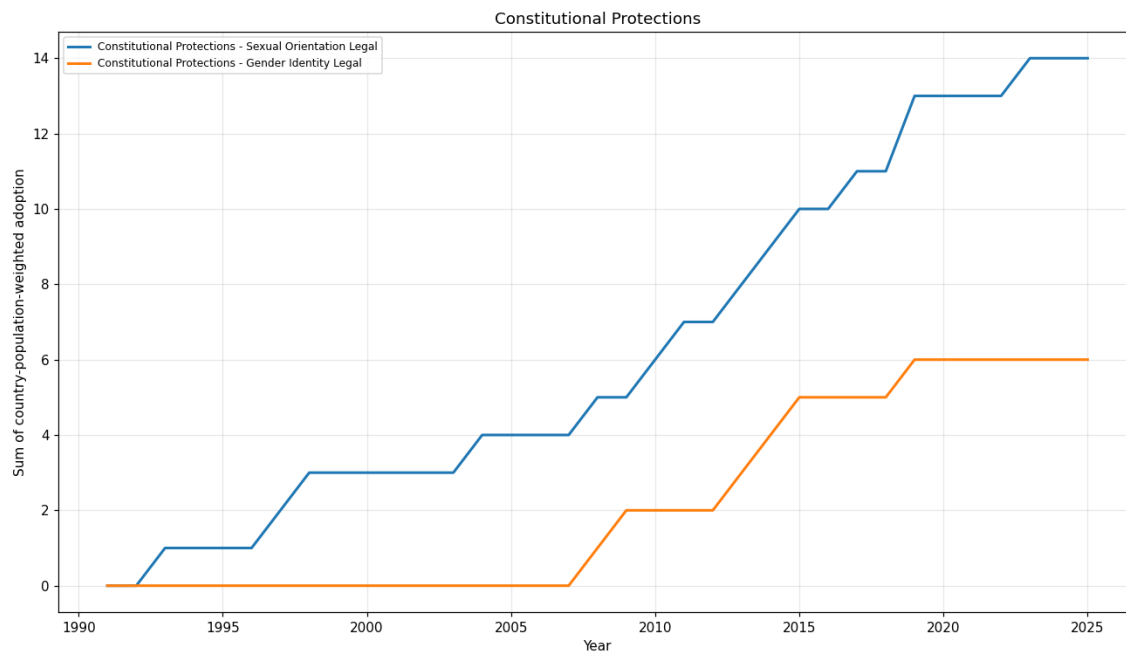


Figure 4.7 — Global trend in constitutional protections, 1991–2025.

Status

Status takes one of: Legal, Illegal, Equal, Unequal, Covered, Restricted. The status label is paired with the Law to specify the direction of the underlying policy. The mapping from Law × Status to substantive orientation (progressive vs. regressive) is in §6.5.

4.3 Outcome variable: Proportion

Proportion captures the fraction of the country's 2000/2001 census population living under the policy in the given year, conditional on the specified Status. For national-only policies, Proportion is binary (0 or 1). For countries with subnational adoption trajectories — currently 18 (Argentina, Australia, Austria, Bosnia and Herzegovina, Brazil, Canada, India, Indonesia, Iraq, Japan, Mexico, Nigeria, Russia, Spain, Switzerland, Tanzania, United Kingdom, United States) — Proportion is continuous and computed as:

Proportion = $\Sigma(\text{2000 census population of subnational units that have adopted by year } Y) / \text{total national 2000 census population}$

Using a constant denominator ensures that year-to-year change reflects policy adoption alone, not demographic change. Subnational units include states, provinces, autonomous communities, and — where a policy operates at that level, as with Japan's partnership-oath systems — municipalities.

In a small number of unitary-state cases, a fractional value instead encodes the documented partial scope of the instrument itself during part of its life (for example, an instrument that was applied to only a subset of its later scope before an amending law). The rationale for any such value is recorded in the group's source chain.

4.4 Temporal metadata

Column	Records
Year_of_National_Adoption (YNA)	Adoption year of the establishing NATIONAL instrument (signing, Royal Assent, promulgation, final court ruling — see §6.1). For purely national instruments this is the adoption year of that statute or decree; for graduated subnational rollouts it is the adoption year of the instrument that completes national coverage. Left empty when the policy never achieves nationwide validity. YNA is the adoption year, not the in-force year; where the establishing statute was signed in a year that precedes its entry into force, YNA can precede the first year Proportion reaches 1.0 (see YNI). For canonical never-criminalized countries (§6.4) YNA is the Correlates of War entry year. For regimes operative before their earliest citable instrument, YNA anchors that instrument (§6.12, documented-instrument anchoring).
Year_of_First_Adoption (YFA)	Year of FIRST adoption — for purely national instruments the same as YNA (signing year); for graduated subnational rollouts the adoption year of the first jurisdiction to adopt. YFA can post-date YNA when signing precedes entry into force AND no earlier subnational adoption exists — in that case YFA aligns with the first year Proportion exceeds 0 (the in-force year of the national instrument).
Year_of_National_Repeal (YNR)	Year the repealing instrument was signed (adoption-year convention, §6.1).
Year_Law_Changed (YLC)	Year of any amending instrument's signing. Semicolon-separated list when several amending instruments mark distinct change-points (e.g., '2019;2020').
Year_National_Implementation (YNI)	Year the establishing instrument actually came into force / had its first practical effect; $\geq$ YNA. Equals YNA when the instrument took effect in its adoption year.

4.5 Supplementary metadata

Blood_Restriction_Period

Length of the MSM blood-donation deferral: 'Permanent', '5-year deferral', '12-month deferral', '6-month deferral', '4-month deferral', '3-month deferral', 'Individual risk assessment', 'No restriction'. Populated only on Blood Donations rows; drives the Ease of Access mapping in §5.1.5.

Max_Punishment

Maximum imprisonment for regressive laws (numeric years) or one of: 'Life', 'Death', 'Admin only', or blank. Where the coded provision contains graduated tiers (a basic and an aggravated form of the same offence), the recorded value is the statutory maximum across those tiers, with the tier structure documented in the group's source chain.

Fine_Amount

Maximum fine in local currency, or 'Yes'/'No' indicating whether fines are part of the penalty. Treat as character.

Sexes_Covered

Numeric Gender Scope score: 1.00 = the statute covers all genders; 0.50 = single-sex coverage (a male-only, female-only, or trans-only instrument). Blank when not applicable. Populated on every active row; the coding convention is in §6.8.

Sub_National_Variation / Sub_National_Coverage

Binary flag (1 = the policy varies by subnational jurisdiction; 0 = otherwise) and a free-text description of the subnational scope, including cases where a separate subnational legal regime exists without changing the national Proportion (e.g., Zanzibar's penal code within Tanzania).

Gender_Change_Requirement

For Gender Marker Change Legal rows, the requirement attached to legal gender recognition in the row's year. The field is time-varying: value follows the operative instrument in each year, and flips at the adoption year of any reforming statute (documented in Year_Law_Changed and the source chain). Permitted values, from most to least accessible:

- Self-ID (1.0) — self-declaration procedure; no medical, psychological, or surgical requirement.
- Court Order (0.5) — judicial procedure retained but no medical/surgical evidence required (e.g., France after Loi 2016-1547, Greece after Nomos 4491/2017).
- Medical Diagnosis (0.5) — medical or psychological certification required; no surgery.
- Medical/Psychological (0.5) — expert declaration of persistent gender identity, typically with a waiting period (e.g., Netherlands after 2014).
- Mixed (Surgery + Medical/Psychological) (0.25) — combination of surgical and medical requirements short of full sterilization.
- Surgery (0.25) — gender-affirming surgery required; no sterilization mandate.
- Surgery+Sterilization (0.0) — surgery and sterilization both required.
- No Legal Route (0.0) — no procedure available; the row typically also carries a populated Year_of_National_Repeal.

Blank where the era's requirement is not specified by the instrument; the Ease of Access derivation treats an unspecified requirement as the 0.5 midpoint. Drives the mapping in §5.1.5.

Civil_Union_Benefits

Numeric 1–5 indicating civil-union benefit scope (1 = minimal registration; 5 = full marriage-equivalent including joint adoption; half-points permitted). Divided by 5 to yield Ease of Access (§5.1.5).

Scope_Detail

Two instruments in the same policy area can look identical on paper but differ substantially in reach — one might apply only to minors while another covers all ages, one might carry criminal penalties while another is administrative, one might cover all official documents while another only passports. Scope_Detail records that within-policy reach as a controlled label (e.g. 'Full_Coverage', 'Minors_Protected_Criminal', 'ID_Or_Passport_Only'). The labels are policy-area-specific — each area has its own vocabulary tailored to the substantively meaningful distinctions in that area. Values are time-varying (per row), so documented scope changes — an amendment extending a minors-only ban to all ages, or a passport-only third-gender marker later extended to all documents — appear as value changes in the year the amending instrument was adopted. Values feed the Ease of Access indicator in the composite index (§5.1.5); a blank cell defaults to the area's presumption (full scope, 1.0). The vocabularies defined for each policy area:

Policy area	Values (Ease of Access value in parentheses)
Gender Affirming Care — Covered	Full_Coverage (1.0) · Regulated_Unfunded (0.75) · Hormones_Only (0.5) · Case_By_Case (0.25)
Gender Affirming Care — Restricted	Total_Ban (1.0) · Modality_Limited (0.5) · Guideline_Restriction (0.25)
Conversion Therapies — Illegal	All_Persons_Criminal (1.0) · Minors_Protected_Criminal (0.75) · Licensed_Professionals (0.5) · Advertising_Or_Partial (0.25)

Employment / Goods–Services Discrimination — Illegal	Full_Civil_Remedy (1.0) · Administrative_Sanctions (0.75) · Indemnity_Or_Dismissal_Only (0.5) · Public_Sector_Only (0.5)
Third-Gender Recognition — Legal	All_Documents (1.0) · ID_Or_Passport_Only (0.5) · Judicial_Individual_Orders (0.25)
Morality/Propaganda — Legal	All_Ages_General (1.0) · Minors_Directed (0.5) · Sector_Limited (0.25)
LGBTQ+ Civil Society Restrictions — Legal	Complete_Ban (1.0) · Registration_Or_Funding (0.5) · Assembly_Only (0.5)

Vocabulary notes: Hormones_Only includes blockers-only regimes for minors; Modality_Limited covers surgery-only or blockers-only restrictions; Licensed_Professionals includes indirect diagnostic prohibitions and licensing-board bans; the Morality vocabulary records the breadth of expression bans (general, minors-directed, or sector-limited).

4.6 Evidence of Enforcement column

Evidence_of_Enforcement (column 29) records documented state enforcement of — or documented access under — the policy in the given year, coded on a five-anchor presumption scale {0.00, 0.25, 0.50, 0.75, 1.00}. The operational definitions are in §6.7. This is the only indicator stored directly as a dataset column because there is no underlying raw field that mechanically determines its value — every entry departing from the presumption rests on a citation in one of the row's Source columns.

The four other indicators that drive the composite index are computed at runtime by the R script from the dataset's raw fields:

Indicator	Scale	Computed from
Population Coverage	[0, 1]	Proportion (population-weighted subnational share)
Gender Scope	{0.00, 0.50, 1.00}	Sexes_Covered (numeric column)
Penalty Severity	[0, 1]	Max_Punishment + Fine_Amount
Ease of Access	[0, 1]	Scope_Detail, Gender_Change_Requirement, Civil_Union_Benefits, Blood_Restriction_Period, or policy-specific defaults

Computing the indicators at runtime rather than storing them eliminates the risk of a stored indicator drifting out of sync with its raw field after an edit.

4.7 Source columns

Twelve source-citation slots per (Country, Law, Status) group:

- Source_1 — primary statute / court ruling / decree citation. See §6.9.
- Source_2 — substantive secondary citation.
- Source_3 through Source_12 — additional citations, including aggregator entries with a country-specific URL or chapter pointer.

All values in every row are attributable to one of the row's Source_1 ... Source_12 columns. There is no separate per-indicator source registry (§6.10).

4.8 Quality columns

data_confidence

Per-group rating of source-triangulation quality:

- high — at least three sources, with Source_1 a primary-instrument citation dated within ± 5 years of YNA
- medium — at least two sources, with Source_1 a substantive citation; or coverage that is interpretive, indirect, or partial in scope (§6.6, §6.6.1)
- low — below the medium bar (rare; typically uncodified-Sharia or thinly documented regimes)

Every substantive group carries a rating.

row_type

- substantive — the row's group has ≥ 1 year where Proportion > 0
- structural_placeholder — group is all-zero across 1991–2025; kept for panel balance only

5. Index Construction

The construction script is titled Velasco_LGBTI_Index_Construction.R and is available at kristophervelasco.com.

5.1 The five indicators

Each policy is measured by up to five indicators, each scored on $[0, 1]$ where higher means the policy is 'more on' for the country in that year. All five are time-varying — values change at documented change-points anchored in the Source_N columns (§6.7).

5.1.1 Population_Coverage

The share of the country's 2000/2001 census population living under the policy in the given year. Equal to the Proportion column. For national-only policies this is binary $\{0, 1\}$; for subnationally varying countries it is a continuous cumulative-adoption share.

Logic: Population_Coverage = Proportion.

5.1.2 Gender_Scope

The scope of genders subject to the law:

Sexes_Covered value	Gender_Scope
1.00 (all genders)	1.0
0.50 (male-only, female-only, or trans-only)	0.5
blank (no law / structural placeholder)	0.0

5.1.3 Penalty_Severity

Used for regressive policies. Higher means harsher penalty.

Max_Punishment	Penalty_Severity
Death penalty	1.0
Life imprisonment	0.8
> 15 years imprisonment	0.6
> 3 and ≤ 15 years	0.4
≤ 3 years	0.2
No law / administrative only	0.0

5.1.4 Evidence_of_Enforcement

Five-anchor variable capturing documented enforcement (regressive policies) or documented access and use (progressive policies) under a presumption framework: 1.00 = operation documented in the era; 0.75 = presumed operative, record silent (the default for event-based policies); 0.50 = documented rare or partial application; 0.25 = documented threats-only or refusal-to-apply; 0.00 = documented dead letter or pre-in-force years. Legality defaults and self-executing regimes carry 1.00 categorically. Forward-fill applies between documented change-points; each change-point requires a Source_N citation referencing the specific event. Full operational definitions are in §6.7.

Documented enforcement sources include Amnesty International country reports, Human Rights Watch World Report, the U.S. State Department Country Reports on Human Rights Practices, ILGA State-Sponsored Homophobia, and LexisNexis / Factiva newspaper archives.

5.1.5 Ease_of_Access

Bidirectional indicator. For progressive policies it captures how readily an LGBTI person can exercise the right; for regressive policies it captures the scope of the restriction. In both directions the factor-orientation step in §5.2 guarantees that final scores are monotone in policy robustness. Derivations for progressive policies:

Policy	Derivation
Gender Marker Change Legal	Self-ID = 1.0; Court Order = 0.5; Medical Diagnosis = 0.5; Medical/Psychological = 0.5; Surgery (incl. mixed surgery regimes) = 0.25; Surgery+Sterilization = 0.0; No Legal Route = 0.0; unspecified = 0.5
Civil Unions Legal	Civil_Union_Benefits / 5 (1-to-5 scale → 0.2–1.0)
Blood Donations Legal	No restriction or individual risk assessment = 1.0; 3-/4-month deferral = 0.75; 6-month = 0.5; 12-month = 0.25; permanent or 5-year = 0.0
Employment / Goods–Services Discrimination Illegal	Scope_Detail remedy depth: Full_Civil_Remedies = 1.0; Administrative_Sanctions = 0.75; Indemnity_Or_Dismissal_Only = 0.5; Public_Sector_Only = 0.5
Conversion Therapies Illegal	Scope_Detail ban scope: All_Persons_Criminal = 1.0; Minors_Protected_Criminal = 0.75; Licensed_Professionals = 0.5; Advertising_Or_Partial = 0.25
Gender Affirming Care Covered	Scope_Detail modality: Full_Coverage = 1.0; Regulated_Unfunded = 0.75; Hormones_Only = 0.5; Case_By_Case = 0.25
Third-Gender Recognition Legal	Scope_Detail document scope: All_Documents = 1.0; ID_Or_Passport_Only = 0.5; Judicial_Individual_Orders = 0.25
Marriage Equality Legal, Joint Adoption Legal, hate-crime and incitement protections, Gender Assignment Surgeries on Children Illegal, Transgender Military Legal	1.0 once active (no within-policy gradient on this axis)

Derivations for regressive policies:

Policy	Derivation
Marriage Equality Illegal	1.0 (a codified opposite-sex marriage definition is treated as fully entrenched)
Gender Affirming Care Restricted	Scope_Detail: Total_Ban = 1.0; Modality_Limited = 0.5; Guideline_Restriction = 0.25; blank = 1.0 presumption
LGBTQ+ Civil Society Restrictions Legal	Scope_Detail: Complete_Ban = 1.0; Registration_Or_Funding = 0.5; Assembly_Only = 0.5
Blood Donations Illegal	Deferral mapping shared with the Legal scale (shorter deferral = higher)

	value); the factor-orientation step in §5.2 aligns the score so that deeper restriction raises the regressive factor
--	--

Ease_of_Access is not part of the indicator set for policies with no within-policy gradient on this axis: Same-Sex Acts (both statuses), Age of Consent (both), Death Penalty, Morality/Propaganda, LGB Military, Religious Exemption Laws, and Constitutional Protections. Where Ease of Access is constant within a policy's pool, the CFA drops the constant indicator automatically (§5.2).

5.2 Confirmatory Factor Analysis

For each policy (Law × Status combination), a one-factor CFA model is estimated pooling all country-year observations where Proportion > 0. Every policy carries at least three loading indicators — the minimum needed to identify a one-factor model — and as many as five when each is conceptually defined for the policy. Indicators with no within-policy variance are dropped (a latent variable cannot be identified from a constant column), and remaining missing indicator values are imputed with the policy-pool mean. Latent variance is fixed at 1 (std.lv = TRUE in lavaan::cfa). Factor scores are min-max normalized to [0, 1] within each policy, and the score direction is reversed whenever the loadings sum is negative, so final scores are always monotone in policy robustness.

For policies whose indicators retain insufficient variance after pooling (typically progressive rights-based policies where Proportion is the only varying indicator), the script falls back to the unweighted indicator mean — mathematically the limiting case of CFA with equal loadings.

5.3 Sample R output

The R output below shows the standardized factor loadings and fit indices for Same-Sex Acts (Illegal) — one of the five policies for which the lavaan CFA fits directly; the others fall back to the indicator-mean limit case (§5.2). In this fit the latent factor emerged oriented toward leniency (negative loadings on the severity indicators); the pipeline automatically reversed the factor scores. Fits for the other four CFA-fitted policies are in Appendix A.

```
> Rscript Velasco_LGBTI_Index_Construction.R

CFA for Same-Sex Acts (Illegal); n = 2773

Standardized factor loadings:
  lhs op                                rhs    est    se      z std.all
policy =~      ind_population_coverage  0.019 0.004  4.88   0.177
policy =~              ind_gender_scope -0.080 0.010 -7.65  -0.336
policy =~              ind_penalty_severity -0.081 0.011 -7.29  -0.298
policy =~ ind_evidence_of_enforcement -0.158 0.020 -7.90  -0.372

Fit indices:
  chisq    df pvalue    cfi    tli  rmsea  srmr
  8.484   2.000  0.014  0.949  0.846  0.034  0.016

[sign of factor scores reversed: sum of loadings < 0]
```

The standardized loadings show that ind_evidence_of_enforcement (|0.372|) and ind_gender_scope (|0.336|) carry the most signal, followed by ind_penalty_severity (|0.298|) and ind_population_coverage (0.177). Countries differ primarily in how strictly criminalization is enforced, whether the ban targets a single sex or all genders, and how harsh the maximum penalty is, with subnational scope contributing a smaller secondary dimension because nearly all criminalizing states apply the ban nationwide.

CFI = 0.949, RMSEA = 0.034, and SRMR = 0.016 indicate acceptable fit (both incremental and residual measures fall within conventional thresholds; TLI = 0.846 remains below 0.90, which is common for four-indicator models with unequal loading magnitudes). The factor scores are used for relative within-policy weighting only, not for structural inference.

5.4 Composite index calculation

$\text{Composite_Index}(c, y) = \sum \text{factor_score}(\text{progressive policies}) - \sum \text{factor_score}(\text{regressive policies})$. The index aggregates 33 Law $\times$ Status combinations — 23 progressive and 10 regressive. The four remaining substantive combinations (Incitement to Hatred Illegal, Joint Adoption Illegal, LGB Military Illegal, Transgender Military Illegal) are excluded by design: each is the direct complement of an on/off policy whose progressive direction is already in the index, so entering both directions would count the same policy space twice. The empirical range is approximately -4.5 to $+18.6$ across the 195-country sample.

5.5 Output

The script writes a single canonical file:

- LGBTI_Composite_Index.csv — Country $\times$ Year panel (6,750 rows) with columns Country, ISO, Year, Progressive_Score, Regressive_Score, Composite_Index, Unweighted_Index.

5.6 Composite index trends, 1991–2025

The figure below shows the global distribution of the composite index across the 195-country sample for every year since 1991. The bold line is the global mean; the dark band is the 25th–75th percentile range across countries; the light band is the 10th–90th. The global mean rose from $+0.24$ in 1991 to $+3.60$ in 2025, with the upper quartile pulling away from the median continuously through the period — global divergence, not convergence, is the dominant pattern.

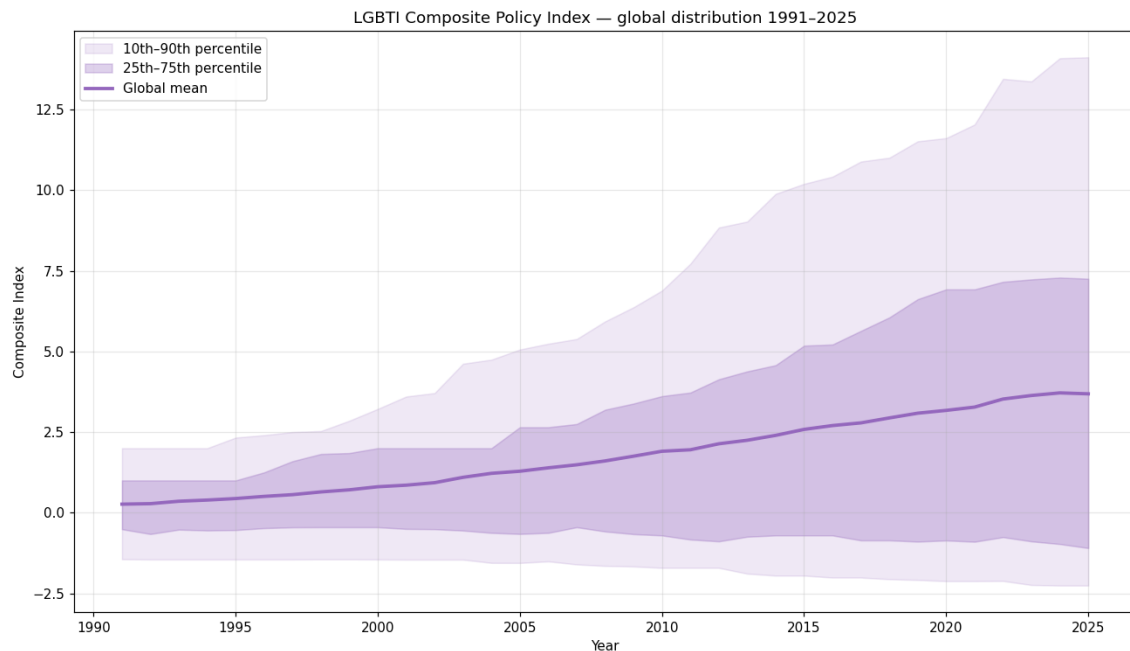


Figure 5.1 — LGBTI Composite Policy Index (33 policy-status combinations), global distribution 1991–2025.

6. Standard Coding Protocol

6.1 Adoption-year convention

The dataset codes the YEAR THE LAW WAS ADOPTED, not when it came into force.

The convention has two layers. First, every instrument is dated to its ADOPTION year — the year the bill was signed / received Royal Assent / the decree was promulgated / the final court ruling was handed down — never its in-force or implementation year (Year_National_Implementation is the separate column for the in-force / effective / first-used date; it is $\geq$ the adoption year whenever both are populated). Second, the two adoption-year columns anchor different points of a policy's territorial trajectory: Year_of_National_Adoption (YNA) is the adoption year of NATIONWIDE validity — operationally, the first year the group's Proportion equals 1.0 — while Year_of_First_Adoption (YFA) is the adoption year of the FIRST instance, operationally the first year Proportion exceeds 0. For a purely national instrument the two coincide. For a policy that never achieves nationwide validity (e.g., state-level conversion-therapy bans in the United States, or the Sharia-state death-penalty provisions in Nigeria), YNA is left empty and YFA carries the anchor.

Worked example — United Kingdom, Incitement to Hatred (sexual orientation): Northern Ireland was the first jurisdiction (Criminal Justice (No.2) (NI) Order 2004, in force 28 September 2004) → YFA = 2004 with Proportion $\approx$ 0.03; England & Wales followed (Criminal Justice and Immigration Act 2008 provisions, in force 23 March 2010) → Proportion $\approx$ 0.92; Scotland completed national coverage (Hate Crime and Public Order (Scotland) Act 2021, in force 1 April 2024) → YNA = 2024 with Proportion = 1.0.

Canonical worked examples:

Country	Group	YNA (adoption)	YNI (implementation)
Bhutan	Same-Sex Acts Legal	2020 (Penal Code Amendment Act passed 10 Dec)	2021 (Druk Gyalpo royal assent 17 Feb)
Fiji	Same-Sex Acts Legal	2009 (Crimes Decree 44/2009 promulgated 5 Nov)	2010 (in force 1 Feb)
Lesotho	Same-Sex Acts Legal	2010 (Penal Code Act 6/2010 enacted)	2012 (in force 9 March)
Australia	Age of Consent Equal	2016 (Qld assented 6 Sept)	2017 (Qld in force 30 Aug)
Switzerland	Civil Unions Legal	2005 (referendum 5 June)	2007 (PartG SR 211.231 in force 1 Jan)
UK	Marriage Equality Legal	2019 (NI Act 2019 c.22 RA 24 Oct)	2020 (NI first marriages 11 Feb)

First-statute principle — YNA tracks the FIRST substantive statute that established the current coding. Subsequent revisions, consolidations, re-codifications, renumberings, or transposition amendments that do NOT change the substantive coding do NOT reset YNA. A revised edition that re-publishes the same criminalization scope keeps the original YNA; a new code that re-enacts existing anti-discrimination protections at the same scope keeps the original YNA. YNA only advances when a new statute substantively changes the policy direction (criminalizes a previously legal act, decriminalizes a previously illegal act, or expands/contracts scope in a way that changes the row's

Status coding). Source_1 cites the establishing instrument, with the current codification noted as a parenthetical or in Source_2. The first-statute principle governs which instrument anchors a given year; the YNA/YFA layer above governs which year of the trajectory each column records. Examples:

Country	Group	YNA	Source_1 form
Italy	Age of Consent Equal	1889	Codice penale Zanardelli 1889 (in force 1890; carried forward through the Codice Rocco 1930 and subsequent revisions)
Armenia	Same-Sex Acts Illegal (pre-2003 era)	1934	Soviet RSFSR Penal Code 1934 §121 (codified domestically as Armenian SSR Penal Code 1961)
Bangladesh	Same-Sex Acts Illegal	1860	Indian Penal Code 1860 §377 (operative in Bangladesh since 1971 independence)
Sri Lanka	Same-Sex Acts Illegal	1883	Penal Code Ordinance No. 2 of 1883 §365 (still operative)

6.2 Civil unions persistence after marriage equality

Marriage equality does NOT automatically demote Civil Union coding to 0. Civil union / partnership statutes remain Legal until the legislation creating them is explicitly repealed by statute. Examples:

- Spain Ley 13/2005 (marriage equality) did NOT repeal the 13 autonomous-community parejas de hecho laws — Spain CU Legal stays at ~0.874.
- Argentina Ley 26.618 (2010 marriage equality) did NOT repeal CABA Ley 1.004 (2002) — Argentina CU Legal stays at ~0.092.
- Mexico marriage-equality reforms did NOT repeal CDMX Sociedad de Convivencia or state Pacto Civil statutes.
- US Obergefell (2015) did NOT repeal state CU/DP statutes in CA, NJ, IL, HI, DC, WA, OR, NV, ME, CO.

6.3 Subnational proportion for federal countries

Proportion at year Y for a subnationally coded country equals the cumulative share of the population in subnational units that have ADOPTED the policy by year Y, using constant 2000/2001 census denominators (§4.3).

6.4 Countries that have never criminalized same-sex acts

Thirty-five countries have never adopted a national criminal statute prohibiting consensual same-sex acts between adults. These countries are designated by their Correlates of War (COW) System Membership entry year as YNA. For each canonical country, YNA = YFA = YNI = COW year, and Source_1 reads: "Never criminalized — YNA = country's Correlates of War (COW) system entry year of independence ([YEAR]). No national criminal statute against consensual same-sex acts between adults has ever been adopted at the federal level (Napoleonic / inherited legal tradition)."

The canonical 35:

Country	COW year	Country	COW year	Country	COW year
Andorra	1993	Equatorial Guinea	1968	North Korea	1948
Argentina	1841	Guatemala	1868	Philippines	1946
Belgium	1830	Haiti	1859	Rwanda	1962
Benin	1960	Indonesia	1949	South Korea	1949

Bolivia	1848	Italy	1860	Taiwan	1949
Brazil	1826	Jordan	1946	Timor-Leste	2002
Cambodia	1953	Laos	1953	Uruguay	1882
Central African Republic	1960	Luxembourg	1920	Vanuatu	1981
Congo	1960	Madagascar	1960	Venezuela	1841
Cote d'Ivoire	1960	Micronesia	1991	Vietnam	1954
Democratic Republic of Congo	1960	Monaco	1993		
Djibouti	1977				
Dominican Republic	1887				
El Salvador	1875				

Indicators on canonical never-criminalized Same-Sex Acts Legal groups: Population_Coverage = 1.0; Gender_Scope = 1.0; Penalty_Severity = blank (not applicable); Ease_of_Access not used; Evidence_of_Enforcement = 1.00 (legality-default convention, §6.7: the legal status is fully effective and freely accessible).

Exception — Argentina. Although Argentina has no national criminal statute against consensual same-sex acts, provincial and municipal police edicts (edictos policiales / códigos de faltas) criminalized "public scandal", "immoral conduct", and same-sex intimacy in most provinces well into the 21st century, and were routinely enforced against LGBTI people. Argentina's Same-Sex Acts Legal group is therefore coded on the subnational trajectory of provincial repeals, with Proportion advancing from an 1841 baseline (Buenos Aires abolition of the sodomy provision inherited from Spain) through the province-by-province elimination of the residual edicts, and YNA anchored to 2012 when the last remaining provincial provisions were struck. Argentina is the only canonical-35 country whose YNA does not equal its COW year for this reason.

Extension to Age of Consent Equal — because the canonical 35 never adopted a separate same-sex criminal statute, their age of consent has applied equally to all sexes and orientations since the country's founding. For these countries, YNA = YFA = YNI = COW entry year also applies to the Age of Consent Equal group, unless a specific later statute introduced an unequal-then-restored regime. Source_1 references the country's founding-era penal code with a parenthetical note that age of consent has been equal since enactment.

6.5 How status labels map to policy direction

Status labels are NOT consistently aligned with progressive vs. regressive across all policies. Three conventions apply:

- Criminalization policies (Same-Sex Acts, Death Penalty, Morality/Propaganda, LGBTQ+ Civil Society Restrictions, Religious Exemption Laws): 'Legal' means the restriction is in effect (REGRESSIVE).
- Anti-discrimination policies (Employment Discrimination, Goods/Services Discrimination, Conversion Therapies, Gender Assignment Surgeries on Children): 'Illegal' means the PROTECTION exists (PROGRESSIVE — the discrimination has been made illegal).

- Hate-crime protections and Incitement to Hatred: 'Legal' means the protection is in force (PROGRESSIVE).

Policy orientation reference:

Law	Status with substantive data	Orientation
Same-Sex Acts	Legal / Illegal	Both substantive
Age of Consent	Equal / Unequal	Equal = progressive, Unequal = regressive
Marriage Equality	Legal / Illegal	Legal = progressive, Illegal = regressive
Civil Unions	Legal	Progressive
Joint Adoption	Legal	Progressive
Employment Discrimination – SO/GI	Illegal	Progressive (ban exists)
Goods/Services Discrimination – SO/GI	Illegal	Progressive (ban exists)
Conversion Therapies	Illegal	Progressive (ban exists)
Hate Crime Protections – SO/GI	Legal	Progressive
Incitement to Hatred	Legal	Progressive
LGB Military / Transgender Military	Legal / Illegal	Legal = progressive, Illegal = regressive
Gender Marker Change	Legal	Progressive
Third-Gender Recognition	Legal	Progressive
Gender Affirming Care – Adults/Minors	Covered / Restricted	Covered = progressive, Restricted = regressive
Blood Donations	Legal / Illegal	Legal = progressive (no restriction), Illegal = regressive
Death Penalty	Legal	Regressive
Morality/Propaganda, LGBTQ+ Civil Society Restrictions, Religious Exemption Laws	Legal	Regressive
Constitutional Protections – SO/GI	Legal	Progressive
Gender Assignment Surgeries on Children	Illegal	Progressive (ban exists)

6.6 What counts as a law for coding purposes

The coding threshold is a **BINDING LEGAL INSTRUMENT**. A policy is coded as in force only where it rests on an instrument that is legally binding and enforceable in the relevant domain:

- **COUNTS:** primary legislation (statutes, codes); constitutional provisions; binding executive orders / decrees with the force of law; binding judicial rulings, including constitutional-court and apex-court interpretations and binding EU-law-conforming interpretation (e.g., gender identity reached through the 'sex' ground per CJEU P v S); and binding regulatory / ministerial instruments that carry legal force.
- **DOES NOT COUNT:** non-binding administrative policy, internal civil-service HR manuals, ministry guidance documents, institutional codes of conduct, professional-society resolutions, or aspirational policy statements; and pure private-law tort doctrine (e.g., a general abusive-dismissal remedy with no enumerated protected ground). A bill that has passed but not entered into force, or a decree struck down before taking effect, does not count until / unless operative.

Interpretive coverage is accepted when it is binding — for example, a gender-identity claim litigated under a 'sex' or 'gender reassignment' ground, or a directly enforceable constitutional non-discrimination clause. Where coverage is interpretive or indirect rather than an explicit enumerated ground, Source_1 records the mechanism and data_confidence is set to medium.

6.6.1 Anti-discrimination protections — domain and scope

Anti-discrimination protections (Employment and Goods/Services, by Sexual Orientation and by Gender Identity) are coded Illegal (i.e., the discrimination is prohibited) only where a binding instrument prohibits discrimination on the relevant ground IN THAT DOMAIN. A statute that covers a ground in employment does not, by itself, establish goods/services coverage, and vice versa.

Year_of_National_Adoption tracks the first binding instrument that establishes coverage of that specific ground-and-domain; an earlier statute covering only a different ground (e.g., race / ethnicity or sex) or only a different domain is not the establishing instrument.

Partial-scope coverage is coded as protection-in-force with a scope note in Source_1, the appropriate Scope_Detail value (§4.5), and data_confidence = medium. This includes: public-sector / executive-branch-only instruments; right-of-admission statutes that reach establishments selling goods and services; protections embedded in a broader sex / gender-equality (CEDAW-implementing) statute via an intersectional or 'multiple-grounds' pathway rather than a standalone SOGI clause; and statutory ombudsman / equality-body mandates that are binding but provide only investigatory (non-judicial) enforcement. Subnational-only coverage in a federal or decentralized state is handled through the population-weighted Proportion (§4.3, §6.3), not by coding national protection where none exists.

6.7 How indicators change over time

All five indicators are time-varying. Within an active group's trajectory, indicator values change only at documented change-points anchored in Source_N citations.

Evidence_of_Enforcement (EoE) is coded under a UNIVERSAL PRESUMPTION OF ENFORCEMENT: a policy on the books is presumed operative unless affirmative counter-evidence shows otherwise. Silence in the record is never counter-evidence — eras with no locatable reports hold the presumption. For access-type rights (third-gender markers, marriage, recognition procedures), EoE measures whether people HAVE ACCESS AT ALL, not usage volume; litigation-gated access counts as access. A law adopted but not yet in force is 0.00 for the pre-in-force years (see Year_National_Implementation); pre-in-force gaps span 1–3 years under the adoption-year convention.

Two coding regimes implement the presumption:

Categorical 1.00 regimes. Two classes carry EoE = 1.00 categorically because their operation involves no documentation uncertainty: (a) legality defaults (Same-Sex Acts Legal, Age of Consent Equal) — legality means no prosecutions by definition; and (b) self-executing administrative regimes (Marriage Equality Illegal, Blood Donations Illegal and Legal, Gender Affirming Care Restricted, LGB/Transgender Military Illegal and Legal, Religious Exemption Laws, Joint Adoption, Conversion Therapy and intersex-surgery bans) — the restriction or permission applies every time it is invoked. Both are downgraded only on documented counter-evidence (de facto persecution despite legality; a regime formally in force but documented as unapplied), cited in a Source_N slot.

Graduated presumption (event-based policies). For policies whose operation consists of events — arrests and prosecutions on the regressive side; uptake, issuance, provision, and application on the

progressive side — EoE encodes both operation and documentation on five anchors. The default is 0.75 (presumed operative); documentation moves the value up or down:

Value	Regressive policies — definition
1.00	Enforcement documented in the era: arrests, prosecutions, convictions, executions, executed sanctions
0.75	Presumed enforced — law in force, era record silent (default)
0.50	Documented as rarely enforced (authoritative rarity characterizations; exceptional cases across long eras)
0.25	Documented official threats/rhetoric without enforcement action
0.00	Documented dead letter: official non-prosecution practice, 'no prosecutions ever' admissions
Value	Progressive policies — definition
1.00	Use documented in the era: any citable instance or statistic of the right being exercised (applies from the documented year forward)
0.75	Presumed accessible — in force, use undocumented (default)
0.50	Documented partial application or structural friction
0.25	Documented refusal-to-apply or systematic obstruction
0.00	Pre-in-force years (adoption-to-force gap of 1–3 years), or documented blocked/revoked access

Counter-evidence standards. Regressive: 0.50 requires an authoritative characterization that the law is rarely enforced; 0.00 requires a documented dead letter (official non-prosecution practice, 'no prosecutions ever' admissions, 'not enforced' findings in country human-rights reporting). Death-penalty layer: death sentences imposed count as 1.00 even if not carried out, and state killings outside formal courts count as application; because capital punishment is exhaustively monitored, the absence of any recorded capital case counts as affirmative never-applied evidence (0.00) — the one area where silence rebuts the presumption. Progressive: 0.50 covers documented partial access failure (implementation gaps, systematic obstruction with a formally open channel, structural gaps such as press-offence immunity); 0.00 covers documented blocked access (a recognition route that produced not one successful case) or documented systematic refusal to enforce (e.g., European Court of Human Rights findings of discriminatory refusal to investigate). Zero recorded uses of a protection does NOT downgrade it — it may simply mean no qualifying incidents; new laws with no application record yet hold the presumption.

TIME-VARYING RULE: a documentation lift applies only FROM the documented year forward; preceding years remain at the 0.75 presumption. Lifts are never backfilled to the start of a policy window.

Sourcing standard: the presumption values (1.00 categorical; 0.75 graduated default) require no citation beyond the policy's primary instrument. Every value departing from the presumption must rest on affirmative evidence cited in one of the group's Source_N slots.

Other indicators:

- Population_Coverage = Proportion (population-weighted share of the country under the policy).
- Gender_Scope: 1.0 = all genders covered; 0.5 = sex-specific (male-only, female-only, or trans-only); 0.0 = no law.
- Penalty_Severity: 1.0 = death; 0.8 = life imprisonment; 0.6 = > 15 years; 0.4 = > 3 and ≤ 15 years; 0.2 = ≤ 3 years; 0.0 = no law or administrative only. Changes when an amending statute changes the penalty (Year_Law_Changed).

- `Ease_of_Access`: bidirectional; derived from `Scope_Detail`, `Gender_Change_Requirement`, `Civil_Union_Benefits`, or `Blood_Restriction_Period` per the §5.1.5 tables. Time-varying through the underlying fields (e.g., a `Scope_Detail` change in the year of an amending instrument).

6.8 Sexes_Covered coding

`Sexes_Covered` is numeric: 1.00 when the instrument covers all genders, 0.50 when it covers one sex or gender category only (a male-only sodomy provision, a female-only provision, or a trans-only instrument). The value describes the CODED instrument itself: where a male-only provision is the operative statute of the group, the group carries 0.50 even if companion provisions elsewhere in the code address other conduct. Single-sex coverage is documented in the source chain (statute article and language). The field is blank where no law exists (structural placeholders).

6.9 Source citation convention

Each row carries up to 12 source citations in `Source_1` ... `Source_12`:

- `Source_1` — primary statute / court ruling / decree citation (statute number, section, gazette / case name and date). Never an aggregator entry.
- `Source_2` — substantive secondary citation: scholarly article, judicial commentary, equality-body report, or dated news event. Avoid aggregator entries where possible.
- `Source_3` onwards — aggregator citations (ILGA, Equaldex, Human Dignity Trust, TGEU, Wikipedia, OSCE) are permitted, but each must point to a specific country / policy page or report chapter.

Not permitted in any slot:

- "ILGA, <https://ilga.org>" (name + homepage, no country pointer)
- "Equaldex" (name alone)
- "Human Dignity Trust, <https://www.humandignitytrust.org>" (name + homepage)

6.10 Tracing every value back to a source

Every value in every row is attributable to one of the `Source_1` ... `Source_12` columns of that row's (Country, Law, Status) group. There is no separate per-indicator source registry. The source columns must collectively cover every documented change-point in the group's trajectory.

6.11 Handling policy reversals

The dataset does not assume monotonic progressive trajectories. When a country revokes a previously existing right, the progressive-status Proportion drops to 0, `Year_of_National_Repeal` is populated, and a dedicated `Source_N` column records the repealing instrument. Boundary convention: the progressive status is coded active through the year BEFORE the repealing instrument's adoption year, and Proportion drops to 0 from the adoption year itself (mirroring the adoption-year convention in the other direction).

Examples: Russia Gender Marker Change (Legal drops from 2024 after the July 2023 ban); Kyrgyzstan Gender Marker Change (drops from 2021); India Gender_Change_Requirement (regresses to Surgery in 2019); Sweden, Finland, and the UK Gender Affirming Care – Minors (Covered → Restricted);

Hungary Employment Discrimination – Gender Identity (Act V of 2025 deleted the gender-identity ground from the 2003 Equal Treatment Act — Proportion 1→0 from 2025); Trinidad and Tobago Same-Sex Acts Legal (the 2018 High Court decriminalization was reversed on appeal 25 March 2025, restoring §§13 and 16 — Proportion 1→0 from 2025). The same mechanism applies to a protection that is later found never to have carried over into the operative successor statute: the protected window ends in the year the successor instrument takes effect.

Where a reversal replaces one regime with its opposite (a blood-donation deferral lifted, or legal gender recognition banned), the dataset uses a two-row pattern: the old status row is trimmed with Year_of_National_Repeal populated, and the opposing status row is activated from the reform's adoption year.

A reversal requires a binding repealing instrument (§6.6). A non-statutory political or administrative event — a policy directive, an education-curriculum change, the dissolution of an enforcement body, or a hostile political climate — does NOT by itself constitute a repeal and does not populate Year_of_National_Repeal; the underlying instrument remains coded as in force, with any practical erosion reflected through Evidence_of_Enforcement and data_confidence rather than through Proportion.

6.12 Conventions for individual policy areas

Conversion Therapies — direct vs. indirect bans

Both direct bans (explicit conversion-therapy prohibition) and indirect bans (mental-health declassification laws prohibiting diagnosing homosexuality as illness) are coded as Illegal. Indirect bans carry '(indirect — mental health declassification)' in Source_1. Indirect bans operate at PROFESSIONAL scope only (they bind licensed clinicians, not all persons); this scope limitation is carried in Scope_Detail (Licensed_Professionals) and reflected in data_confidence.

Blood Donations — all deferrals coded as Illegal

Any MSM blood-donation restriction is coded Illegal. The substantive distinction is in Blood_Restriction_Period, not Status. Only individual risk assessment with no MSM-specific deferral is coded Legal.

Gender Affirming Care — functional standard

Both Covered and Restricted are anchored under a FUNCTIONAL standard, applied symmetrically: an instrument qualifies if it operationally determines public provision or restriction of care — statutes, decrees, ministerial circulars, health-authority guidelines and recommendations that define the publicly funded service range, reimbursement and commissioning decisions, medical-council regulations, and binding insurance case law. Documented public-health-system practice eras (a national clinic operating within the public system before the first formal instrument, or a practice-level tightening acknowledged by the health ministry) are retained in the active window with data_confidence = low. Absence of public funding alone does NOT qualify as Restricted, and a clinic's mere existence does not anchor YNA — the first functional instrument does.

Religious Exemption Laws — scope

Codes only broad exemptions for individuals / businesses / organizations to refuse services to LGBTQ+ people. Routine clergy-only exemptions are NOT counted.

Military Service — explicit policy required

LGB Military and Transgender Military are coded only for countries with explicit policies. De facto tolerance or silence is not coded as either Legal or Illegal.

United States Marriage Equality — independent coding

US Marriage Equality Legal and Illegal are independently coded from actual state-by-state policy using constant 2000 census denominators. Some states had neither legalized nor banned same-sex marriage in some years, so Legal + Illegal < 1 in 1995–2012.

Marriage Equality Illegal — explicit codification rule

Marriage Equality Illegal is substantive (Proportion = 1) for any country whose national marriage statute, family code, or constitution explicitly codifies marriage as a union of "one man and one woman" (or equivalent opposite-sex-only language), regardless of when that codification occurred. Source_1 cites the operative statute and its article. Marriage Equality Illegal is a structural placeholder only for countries with no such codification (e.g., common-law jurisdictions where marriage is defined purely by judicial precedent without a codified opposite-sex requirement).

United States Gender Marker Change — subnational coding and the federal layer

US Gender Marker Change is coded subnationally: state birth-certificate amendment regimes (operating since the 1960s, with most states permitting by the 1990s) drive the Proportion series, with the federal passport policy (US Department of State, 9 June 2010, 7 FAM 1320, eliminating the surgery requirement) as the federal-layer anchor. Executive Order 14168 (20 January 2025) ended federal gender-marker changes (Supreme Court stay of the injunction, 6 November 2025); state-level regimes continue in most states, so the 2025 Proportion reflects the share of states still permitting (~0.70) rather than zero.

Gender Marker Change — origin anchoring

Gender Marker Change rows anchor the FIRST binding legal-gender-recognition instrument of any kind, including medicalized and surgery-conditional regimes; later self-identification reforms are documented in the source chain but do not reset YNA. The Gender_Change_Requirement field and the Ease of Access indicator carry the medicalized-to-self-ID gradient. Operative, consistently applied court practice counts as a binding pathway (e.g., declaratory-action practice where a statute is absent); a statute whose implementing legislation was never adopted and under which no changes occur — a dead letter — does not.

Gender identity grounds — explicit enumeration

A gender-identity row (hate crime, incitement, anti-discrimination, constitutional protection) requires gender identity to be explicitly enumerated in the instrument, or covered by binding interpretive doctrine (e.g., gender identity read into the 'sex' ground by an apex court or the CJEU, per §6.6). Residual catch-all clauses ('any other personal circumstance') do NOT qualify.

Judicial recognition routes

Where recognition (third-gender markers, adoption access) exists only through individual court orders rather than a general statute or regulation, the route is coded as in force from the first successful order, with `data_confidence` reflecting the route's consistency: consistently granted, repeatable routes carry medium confidence; isolated orders carry low confidence. The absence of an enabling statute is always noted in `Source_1`.

Documented-instrument anchoring

Where a policy regime was operative at the start of the panel but the earliest citable instrument is later — a blood-donation deferral operating before its first written service framework, a military exclusion practiced before its first codified directive, a sharia-based or colonial-era regime later restated in a modern code, a publicly funded care pathway operating before its first formal guideline — the `Proportion` series reflects the regime's operative status from panel start, and `Year_of_National_Adoption` anchors the earliest citable national instrument, recorded in `Source_1`. Estimated dates for uncitable predecessor instruments are not used. This parallels the *Correlates-of-War* anchoring for never-criminalized countries (§6.4): the anchor is always the earliest citable legal fact.

6.13 Handling transition years when laws change mid-year

When a law changes mid-calendar-year, the dataset does not encode the month. Both statuses may show `Proportion > 0` for that calendar year. Single-year overlap is acceptable; multi-year overlap occurs only in the federal-split and policy-layering cases described in §3.4.

7. Unweighted Index

Alongside the CFA-based composite index (§5), the dataset publishes an Unweighted Index as a simple raw count: the total number of progressive national laws in force minus the total number of regressive national laws in force. It takes no indicator weighting, no subnational variation, and no within-policy variation into account.

Definition: for each Country × Year, a Law × Status combination contributes +1 if it is progressive and `Proportion = 1`, and −1 if it is regressive and `Proportion = 1`. Combinations with `Proportion < 1` contribute 0, even when the policy is partially in force subnationally.

The Unweighted Index is stored as the seventh column of `LGBTI_Composite_Index.csv`.

7.1 Range and trajectory

Empirical range across the 195-country panel, 1991–2025: −6 to +21. The global mean rose from −0.23 in 1991 to +3.77 in 2025. Top of the distribution in 2025: Malta (+21), Germany (+20), Netherlands (+19), Portugal (+19), Spain (+19). Bottom of the distribution in 2025: Brunei (−6), Malaysia (−6), Oman (−6), Nigeria (−5), Saudi Arabia (−5).

7.2 Relationship to the CFA-based Composite Index

The two indices are tightly correlated across the 195-country panel. The annual Pearson correlation rose from $r = 0.959$ in 1991 to $r = 0.993$ in 2025, with a mean of $r = 0.975$ over the full period.

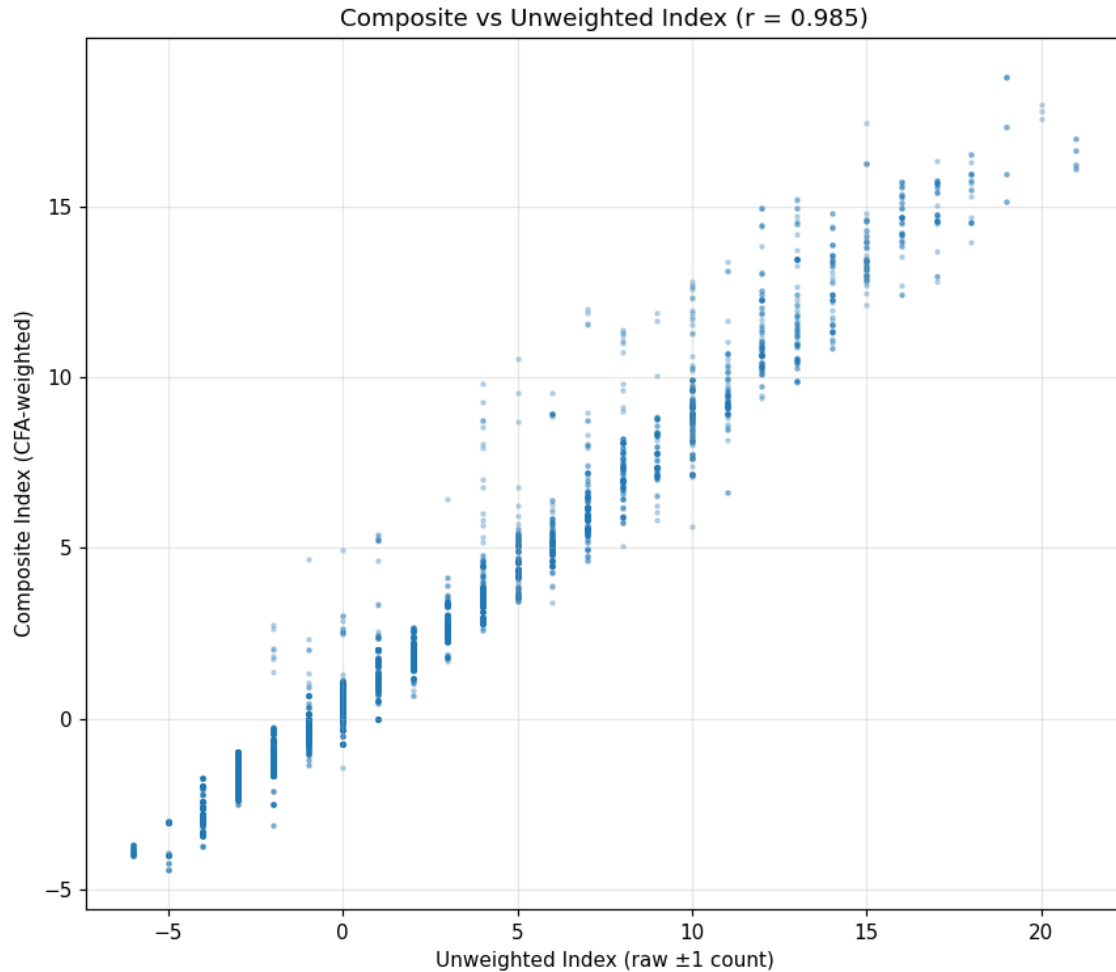


Figure 7.1 — Annual Pearson correlation between the CFA-based Composite Index and the Unweighted Index across the 195-country panel, 1991–2025.

Substantively, the two indices rank countries almost identically. The variance they do not share is informative rather than noise: it is the additional content the CFA-based Composite captures about subnational rollout (via population coverage), penalty severity, within-policy scope, and enforcement that the raw count discards — most visible in countries with active subnational variation or high-intensity regressive enforcement.

8. Sources & Provenance

8.1 Primary data sources

- Velasco (2023) LGBTI Policy Index — foundational dataset.

- ILGA World State-Sponsored Homophobia Report and Trans Legal Mapping Report.
- Human Dignity Trust country profiles.
- Equaldex.
- Trans Rights Indicator Project (TRIP) by Williamson.
- Transgender Europe (TGEU).
- Movement Advancement Project — US state-level LGBTQ+ policy.
- Hague Centre for Strategic Studies LGBTQ+ Military Index.
- Amnesty International, Human Rights Watch, U.S. State Department Country Reports.
- LexisNexis and Factiva newspaper archives.
- National legislative databases (Légifrance, Bundesgesetzblatt, BOE, STF, OSF, gov.uk, etc.) in 10+ languages.
- National census bureaus — 2000/2001 census population data.

8.2 Provenance principle

Every cell in every row is traceable to one of the row's Source_1 ... Source_12 columns. The Source_N columns must collectively cover every documented change-point in the group's trajectory. There is no separate per-indicator or per-event source registry; the row's source columns ARE the registry.

9. Known Limitations

- North America contains only 5 countries (US, Canada, 3 Caribbean states). Researchers may merge with Latin America depending on analytic needs.
- 17 of 54 Law × Status combinations are globally all-zero structural placeholders. Use row_type to filter them out.
- Some country-year-law cells show both paired statuses simultaneously > 0. These reflect legitimate federal subnational splits or single-year transitions where the law changed mid-calendar-year (§3.4, §6.13).
- Status labels for anti-discrimination protections are inverted relative to criminalization policies; see §6.5.
- Fine_Amount mixes numeric values with Yes/No strings; treat as character.
- Year_of_National_Adoption involves coder judgment within ±2 years when the source statute distinguishes parliamentary passage from Royal Assent from gazette publication.
- Evidence_of_Enforcement values at the graduated default (0.75) reflect a presumption rather than documentation; researchers requiring documented-only enforcement can filter to values of 1.00 and below 0.75, each of which carries a citation (§6.7).
- Three coding decisions involve substantive interpretive judgment that researchers may wish to revisit: Kenya Third-Gender Recognition (2022 Children Act, intersex-specific); Pakistan Goods/Services Discrimination – Gender Identity (2018 statute partly struck by Federal Shariat Court 2023); Nepal Constitutional Protections – Gender Identity (Article 42 'gender and sexual minorities' inclusive interpretation).

Appendix A. CFA fit details for all five lavaan-fitted policies

The composite index (\$5) fits a one-factor CFA per policy where indicators have sufficient variance; the remaining policies fall back to the unweighted indicator mean (\$5.2). The table below reports the standardized factor loadings and model-fit indices for every policy where the lavaan CFA succeeded. Chi-square df of 0 corresponds to a saturated three-indicator model, for which the standard fit indices are not defined; loadings are still identifiable.

Policy	n	Indicator	Std. loading	chi ²	df	CFI	TLI	RMSEA	SRMR
Same-Sex Acts (Illegal)	2,773	Population Coverage	0.177	8.484	2	0.949	0.846	0.034	0.016
		Gender Scope	−0.336						
		Penalty Severity	−0.298						
		Evidence of Enforcement	−0.372						
Gender Marker Change (Legal)	1,876	Population Coverage	0.240	0	0	1.000	1.000	0.000	0.000
		Ease of Access	−0.722						
		Evidence of Enforcement	−0.261						
Employment Discrimination – Gender Identity (Illegal)	681	Population Coverage	0.108	0	0	1.000	1.000	0.000	0.000
		Ease of Access	0.313						
		Evidence of Enforcement	0.609						
Third-Gender Recognition (Legal)	193	Population Coverage	0.107	0	0	1.000	1.000	0.000	0.000
		Ease of Access	0.418						
		Evidence of Enforcement	−0.231						
Conversion Therapies (Illegal)	191	Population Coverage	0.462	2.586	2	0.911	0.733	0.039	0.035
		Penalty Severity	0.069						
		Ease of Access	0.267						
		Evidence of Enforcement	0.367						

All other substantive Law × Status combinations (28 progressive, 5 regressive) use the indicator-mean fallback because after imputation and variance-screening they retain fewer than three varying indicators — typically because Proportion is the only meaningful within-policy source of variation. The score is then the min-max-normalized unweighted mean of the available indicators, which is mathematically the limiting case of a CFA in which all loadings are held equal.

Appendix B. Update from v1.0

Version 1.0 is the LGBTI Policy Index published with Velasco (2023) in the American Journal of Sociology: a wide-format country–year matrix of 192 countries × 1991–2019 × 21 binary policy indicators, aggregated into a composite index by the same one-factor CFA methodology used here.

Version 2 keeps that composite construction and rebuilds the release around it to widen its analytic uses and to make every coding decision traceable and reproducible.

- Wider coverage. 195 countries through 2025; 27 policy areas × 54 Law × Status combinations, up from 21 binary items — new areas capture gender-affirming care, blood donation, civil-society restrictions, religious exemptions, and split SO/GI protection tracks.
- Different format. Long-format balanced panel with structural placeholders, so the data drops directly into panel-regression, event-study, and matching frameworks that the v1.0 wide matrix did not support.
- Sub-national trajectories. Federal and devolved countries are coded as population-weighted Proportion series using constant 2000/2001 census weights, rather than the v1.0 binary national flag — enabling within-country variation to be used analytically.
- Codified coding protocol. Every value is re-derived under the §6 protocol (adoption-year and first-statute conventions, binding-instrument threshold, reversal handling, enforcement presumption, documented-instrument and Correlates-of-War anchoring). Individual codings may differ from v1.0 — most commonly because v2.0 dates adoption rather than entry into force.
- Cell-level sourcing. Every cell is traceable to one of twelve per-group source-citation slots, and each group carries a data-confidence rating. v1.0 carried no cell-level sourcing.
- Wider composite scope. The one-factor CFA composite that already anchored v1.0 is now applied across 33 policy-status combinations (up from 21), so newly added areas — gender-affirming care, blood donation, civil-society restrictions, religious exemptions, and split SO/GI protections — feed the composite directly. The subnational Proportion series enters as the population-coverage indicator for federal and devolved countries, so within-country rollout now contributes to the score rather than being flattened to a national flag. An unweighted raw-count index is published alongside as a transparency baseline.

Versioning scheme: minor updates increment the decimal (2.1, 2.2, ...); major revisions increment the integer (3.0).

Version 2.2 (July 2026)

Point update converting the Gender_Change_Requirement field on 23 Gender Marker Change (Legal) groups from a static value to a per-year value that flips at the adoption year of each reforming instrument. The affected groups previously carried a single class across 1991–2025 even where the source chain documented a class-changing reform. This release records the era boundaries directly in the panel:

- Denmark 2014, Colombia 2015, Malta 2015, France 2016, Greece 2017, Brazil 2018, Portugal 2011 and 2018, Iceland 2019, New Zealand 2021 — earlier surgery-based or medical-diagnosis regimes replaced by self-declaration procedures (Court Order class for France and Greece, which retained a judicial requirement without medical evidence).
- Austria 2009, Cyprus 2022, Japan 2023, Luxembourg 2018, Netherlands 2014, Sweden 2013, Ukraine 2016, Croatia 2013, Serbia 2018, Uruguay 2009 and 2018 — additional class-changing reforms of the same type, identified in a systematic sweep and given the same era-boundary treatment.

- India 2019 — regression rather than liberalization: the NALSA v Union of India judgment (2014) established a self-identification right, and the Transgender Persons (Protection of Rights) Act 2019 reintroduced a District Magistrate certificate and, for a binary male/female marker, a medical review that functions as a surgery requirement.
- Canada 2017 — federal completion of a subnational rollout to self-declaration (Ontario's HRTO ruling in XY v Ontario, 2012, dropped the surgery prerequisite; provincial rollout 2012–2016; Immigration, Refugees and Citizenship Canada X-marker in 2017).
- Bulgaria 2023 and Kyrgyzstan 2020 — closure rather than liberalization: the Bulgarian Supreme Court of Cassation issued a binding interpretive decision in 2023 that ruled Bulgarian law does not permit sex-marker change, and Kyrgyzstan withdrew the 2017 Form 48/y administrative procedure in 2020. Both groups now carry Year_of_National_Repeal at those years and Proportion drops to zero afterward.

A Court Order class was added to the Gender_Change_Requirement vocabulary to capture instruments that retain a judicial procedure but require no medical, psychological, or surgical evidence (France post-2016 and Greece post-2017). It maps to 0.5 on the Ease of Access scale, the same as Medical Diagnosis.

The dataset was already time-varying on Gender_Change_Requirement in principle — 15 country groups already flipped their value at reform years before this release — so this update brings the remaining 23 groups into line with the same rule rather than changing the convention. No YNA, YFA, or Proportion values are altered outside the two closure cases above.

Also in this release: the Status label on Mexico and Venezuela's Incitement to Hatred rows was corrected. Both countries had the substantive protection (Mexico's LFPED 2014; Venezuela's Ley Constitucional contra el Odio 2017) filed under Status = Illegal instead of Status = Legal, the convention used by the other 48 countries that criminalize LGBTI-directed incitement. The Status labels were swapped across all 1991–2025 rows for these two country groups; no source, year, or value fields were altered. Mexico's composite rises by +0.71 from 2014 onward and Venezuela's by +0.86 from 2017 onward; the global mean composite in 2025 moves from +3.69 to +3.60.